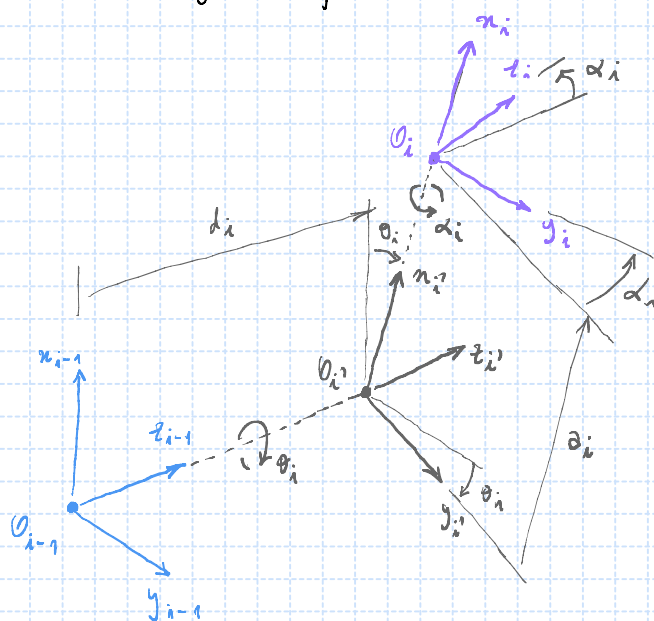


(1) Denavit-Hartenberg transformation



Link i $d_i \theta_i a_i \alpha_i$

Link i is the transformation from frame i-1 to frame i

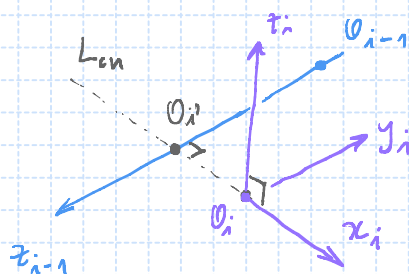
$$A_{\beta}^{\alpha} = \begin{pmatrix} R_{\beta}^{\alpha} & r_{\alpha, \beta}^{\alpha} \\ 0^T & 1 \end{pmatrix}$$

$$A_{\beta}^{\alpha} = A_{\sigma}^{\alpha} A_{\beta}^{\sigma}$$

$$A_{\beta}^{\alpha} \begin{pmatrix} r_{\beta, \sigma}^{\beta} \\ 1 \end{pmatrix} = \begin{pmatrix} R_{\beta}^{\alpha} r_{\beta, \sigma}^{\beta} + r_{\alpha, \beta}^{\alpha} \\ 1 \end{pmatrix} = \begin{pmatrix} r_{\alpha, \beta}^{\alpha} + r_{\beta, \sigma}^{\alpha} \\ 1 \end{pmatrix} = \begin{pmatrix} r_{\alpha, \sigma}^{\alpha} \\ 1 \end{pmatrix}$$

$$A_{\beta}^{\alpha} \begin{pmatrix} r_{\beta, \sigma}^{\beta} \\ 1 \end{pmatrix} = \begin{pmatrix} r_{\alpha, \sigma}^{\alpha} \\ 1 \end{pmatrix}$$

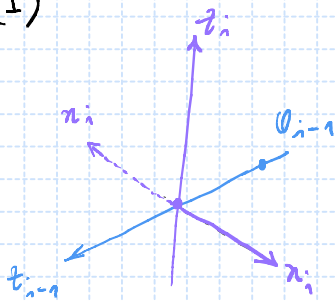
D-H frames construction,



1. Draw all of z_i 's according to the joint type
2. Define origin of frame 0 (O_0)
3. Iterate along i for known z_{i-1}, z_i, O_{i-1}
 - (i) locate common normal (L_{cn}) between z_{i-1} and z_i
 - (ii) locate O_{i-1} and O_i
 - (iii) Draw x_i along L_{cn} , from O_{i-1} to O_i
 - (iv) complete $y_i = z_i \times x_i$

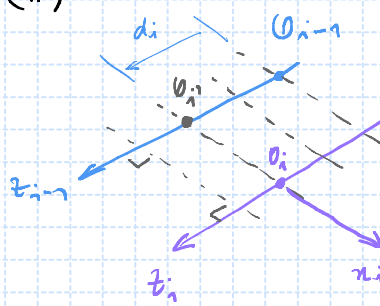
Spiral cases:

(I)



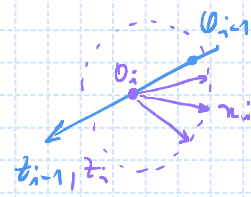
choose x_i normal to the plane formed by z_i and z_{i-1}

(II)



choose O_i along z_i

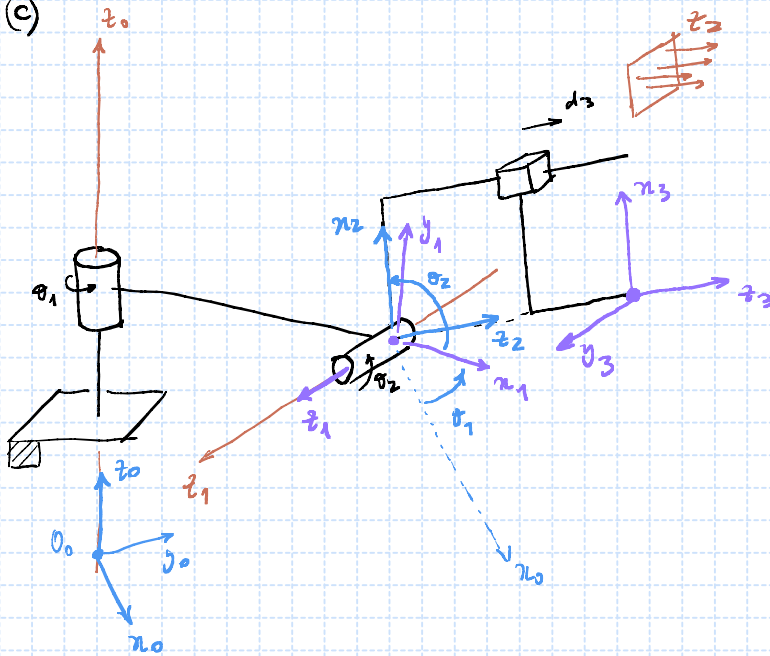
(III)



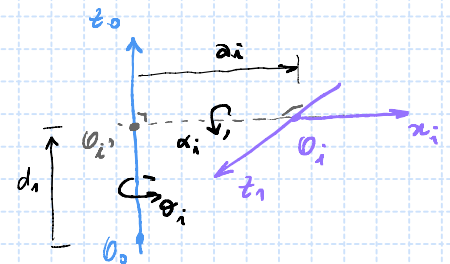
choose O_i along z_i ,
choose x_i perpendicular to z_i

common choice for O_i is the one for which $d_i = 0$.

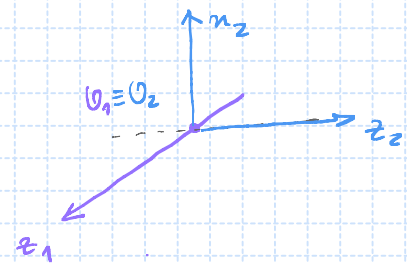
(2) (c)



Link 1 $d_1 \theta_1(t) a_1 \pi/2$



Link 2 $0 \theta_2(t) 0 \pi/2$



Link i	d_i	θ_i	a_i	α_i
1	d_1	θ_1	a_1	$\pi/2$
2	0	θ_2	0	$\pi/2$
$n=3$	d_3	0	0	0

n degrees of freedom

Direct Kinematics (Systematic Method), $A_n^0 = A_1^0 A_2^1 \dots A_n^{n-1}$

$$A_1^0 = \begin{pmatrix} R_1^0 & r_{0,1}^0 \\ 0^T & 1 \end{pmatrix} = \begin{pmatrix} c_1 & 0 & s_1 & a_1 c_1 \\ s_1 & 0 & -c_1 & a_1 s_1 \\ 0 & 1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad A_i^{i-1} = \begin{pmatrix} c_{\theta_i} & -s_{\theta_i} c_{\alpha_i} & s_{\theta_i} s_{\alpha_i} & a_i c_{\theta_i} \\ s_{\theta_i} & c_{\theta_i} c_{\alpha_i} & -c_{\theta_i} s_{\alpha_i} & a_i s_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$A_2^1 = \begin{pmatrix} R_2^1 & r_{1,2}^1 \\ 0^T & 1 \end{pmatrix} = \begin{pmatrix} c_2 & 0 & s_2 & 0 \\ s_2 & 0 & -c_2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

Link i	d_i	θ_i	α_i	a_i
1	d_1	θ_1	0	$\pi/2$
2	0	θ_2	0	$\pi/2$
3	d_3	0	0	0

$$A_3^2 = \begin{pmatrix} R_3^2 & r_{2,3}^2 \\ 0^T & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

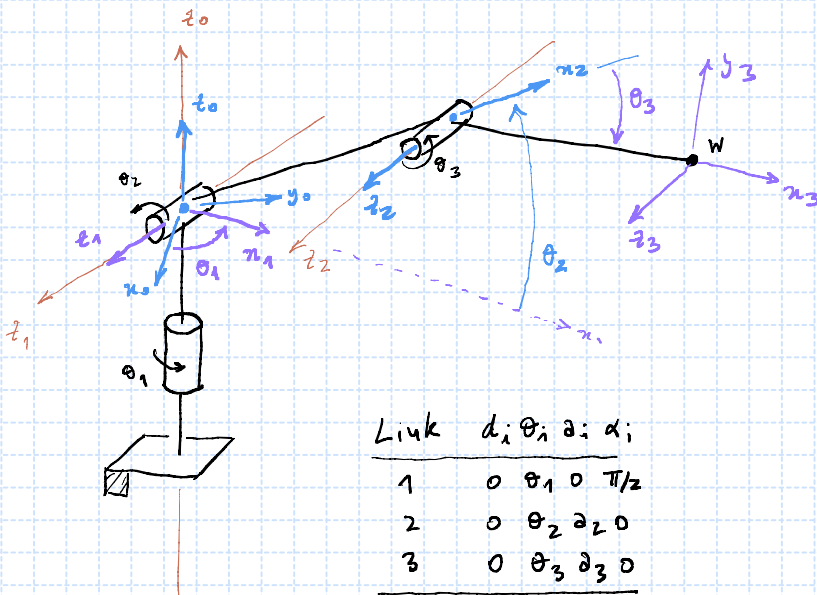
$$A_3^0 = A_1^0 A_2^1 A_3^2 = A_2^0 A_3^2$$

$$A_2^0 = \begin{pmatrix} c_1 c_2 & s_1 & c_1 s_2 & a_1 c_1 \\ s_1 c_2 & -c_1 & s_1 s_2 & a_1 s_1 \\ s_2 & 0 & -c_2 & d_1 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

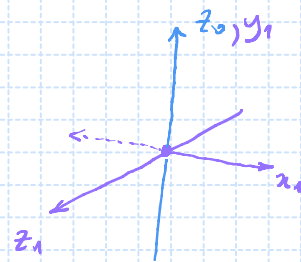
$$A_3^0 = A_2^0 A_3^2 = \begin{pmatrix} c_1 c_2 & s_1 & c_1 s_2 & d_3 c_1 s_2 + a_1 c_1 \\ s_1 c_2 & -c_1 & s_1 s_2 & d_3 s_1 s_2 + a_1 s_1 \\ s_2 & 0 & -c_2 & -d_3 c_2 + d_1 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} R_3^0 & r_{q3}^0 \\ 0^T & 1 \end{pmatrix}$$

direct kinematics

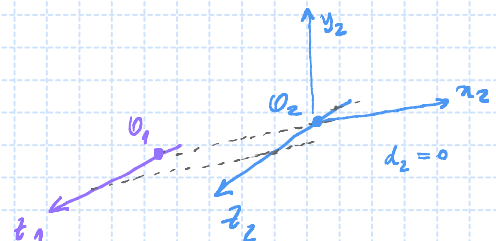
5.



Link 1 $0 \theta_1 0 \pi/2$



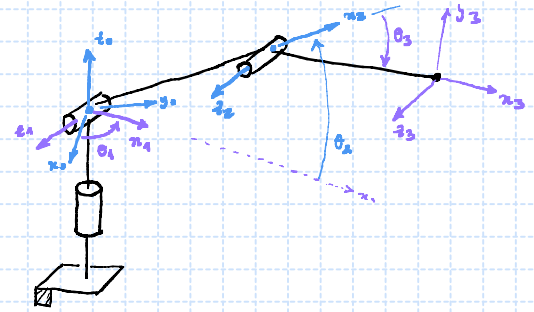
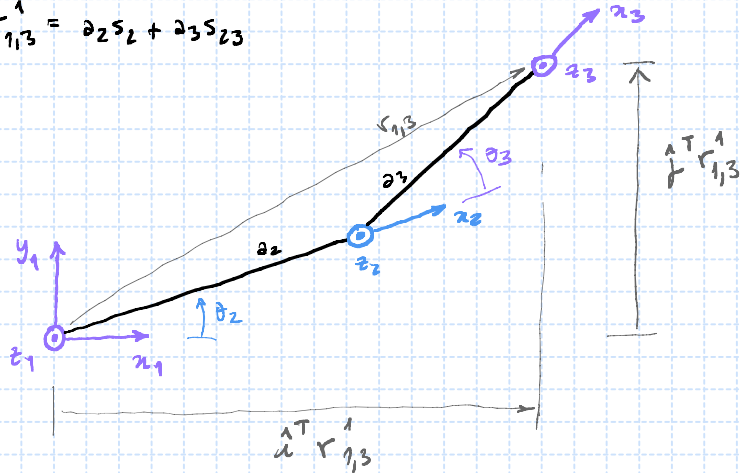
Link 2 $0 \theta_2 0 0$



Direct Kinematics (Geometric Method)

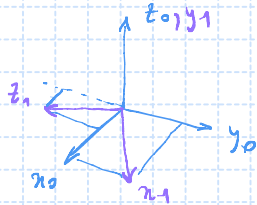
$${}^1_A r_{1,3}^1 = a_2 c_2 + a_3 c_{23}$$

$${}^1_A r_{1,3}^1 = a_2 s_2 + a_3 s_{23}$$



$$r_{1,3}^1 = \begin{pmatrix} a_2 c_2 + a_3 c_{23} \\ a_2 s_2 + a_3 s_{23} \\ 0 \end{pmatrix}$$

$$r_{0,3}^0 = r_{0,1}^0 + R_1^0 r_{1,3}^1$$



$$R_1^0 = \begin{pmatrix} c_1 & 0 & s_1 \\ s_1 & 0 & -c_1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$r_{0,3}^0 \equiv p_3 = \begin{pmatrix} (a_2 c_2 + a_3 c_{23}) c_1 \\ (a_2 c_2 + a_3 c_{23}) s_1 \\ a_2 s_2 + a_3 s_{23} \end{pmatrix}, \text{ position of the wrist.}$$